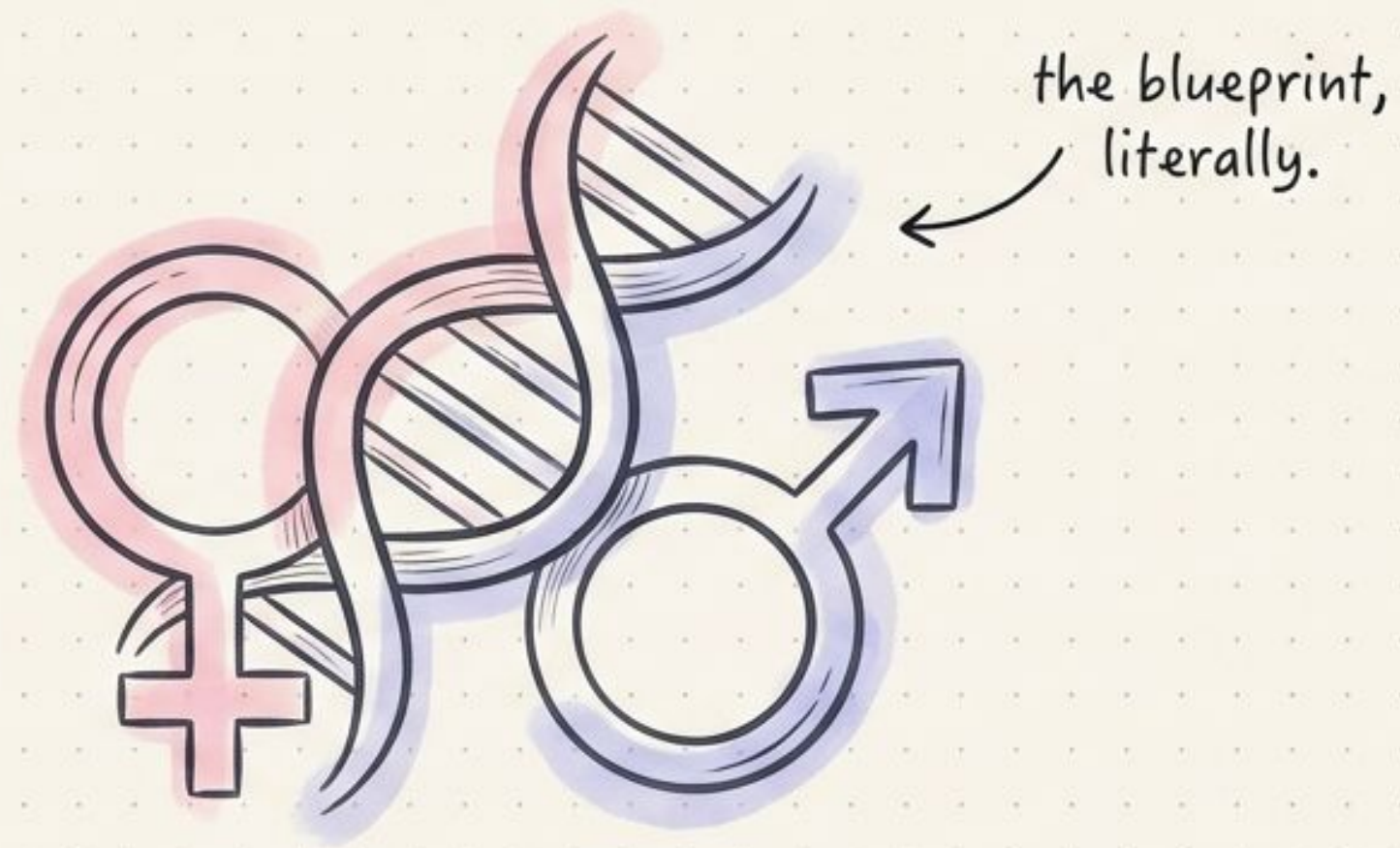




# Main Character Biology

## Anatomy & Physiology of Reproduction

The definitive hardware & software guide to the human reproductive system.

# Quest Log: What We're Unlocking Today

- ☒ **Differentiate the female and the male reproductive systems** (aka checking the hardware specs).
- ☒ **Understand the basis for physiologic processes in females and males** (aka learning the software mechanics).



# The Lexicon

*Let's get the terminology straight, no cap.*

## Reproduction

The process of producing offspring.

## Genitals

External sex organs.

## Primary Sex Characteristics

Sex characteristics that are present at birth.

## Secondary Sex Characteristics

Sex characteristics that emerge during puberty (the ultimate glow-up).



# The Outer Hardware: Female External Genitalia (The Vulva)

## Inventory Notes

**Mons veneris:** Pads of fatty tissue between pubic bone and skin.

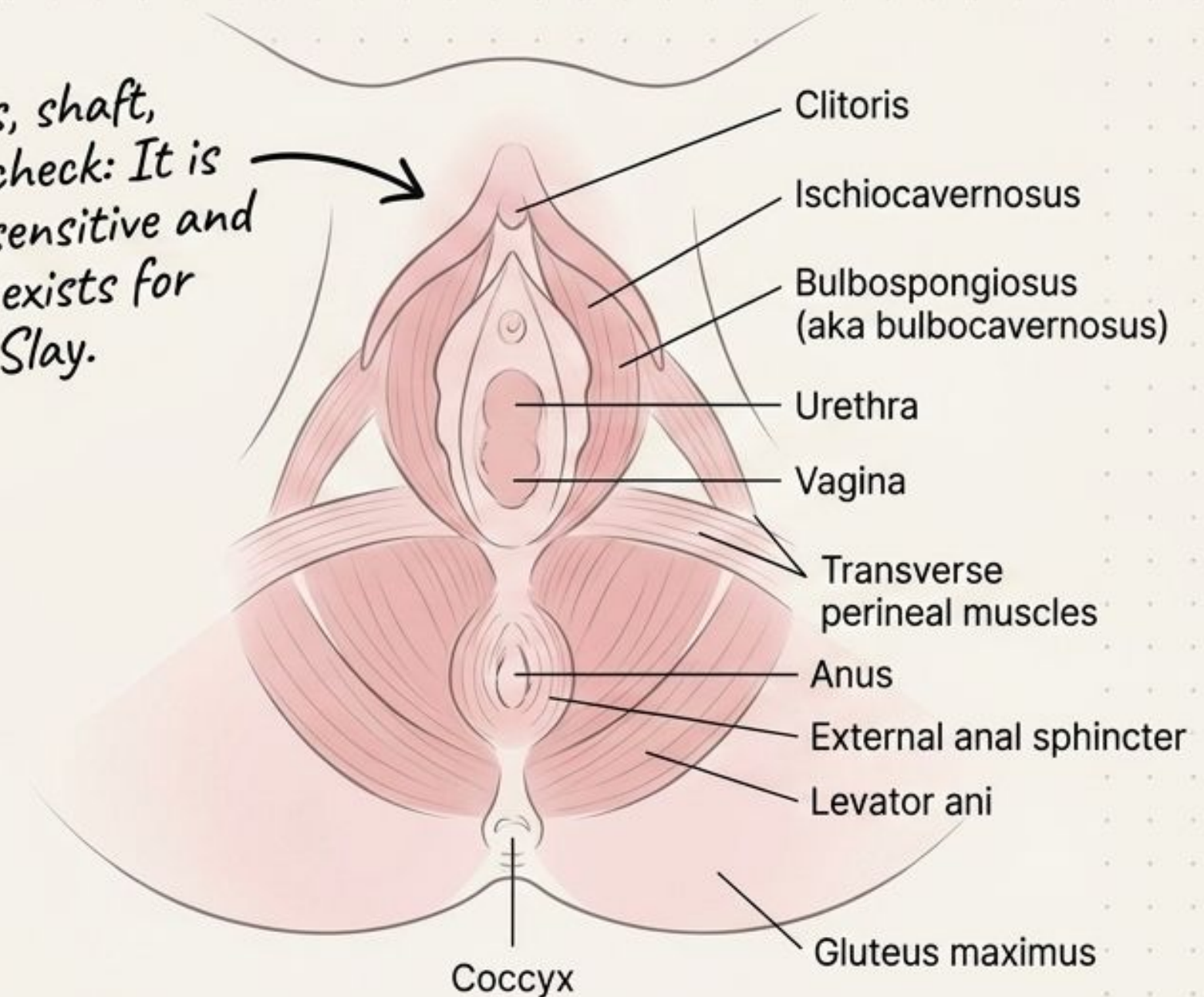
**Labia majora & minora:** Outer lips / inner lips surrounding the vestibule (packed with sweat/oil glands, blood vessels, nerve endings).

**Prepuce:** Clitoral hood.

**Vestibule & Openings:** Area surrounding urethral opening (urination) and vaginal opening (introitus).

**Perineum:** Skin separating genitalia from anus (distance is less in females than males).

*Clitoris (glans, shaft, crura). Vibe check: It is particularly sensitive and literally only exists for stimulation. Slay.*



# The Inner Workings: VIP Access to the Female Tract

**Ovaries:** The powerhouses. Produce estrogen, progesterone, and house ova.

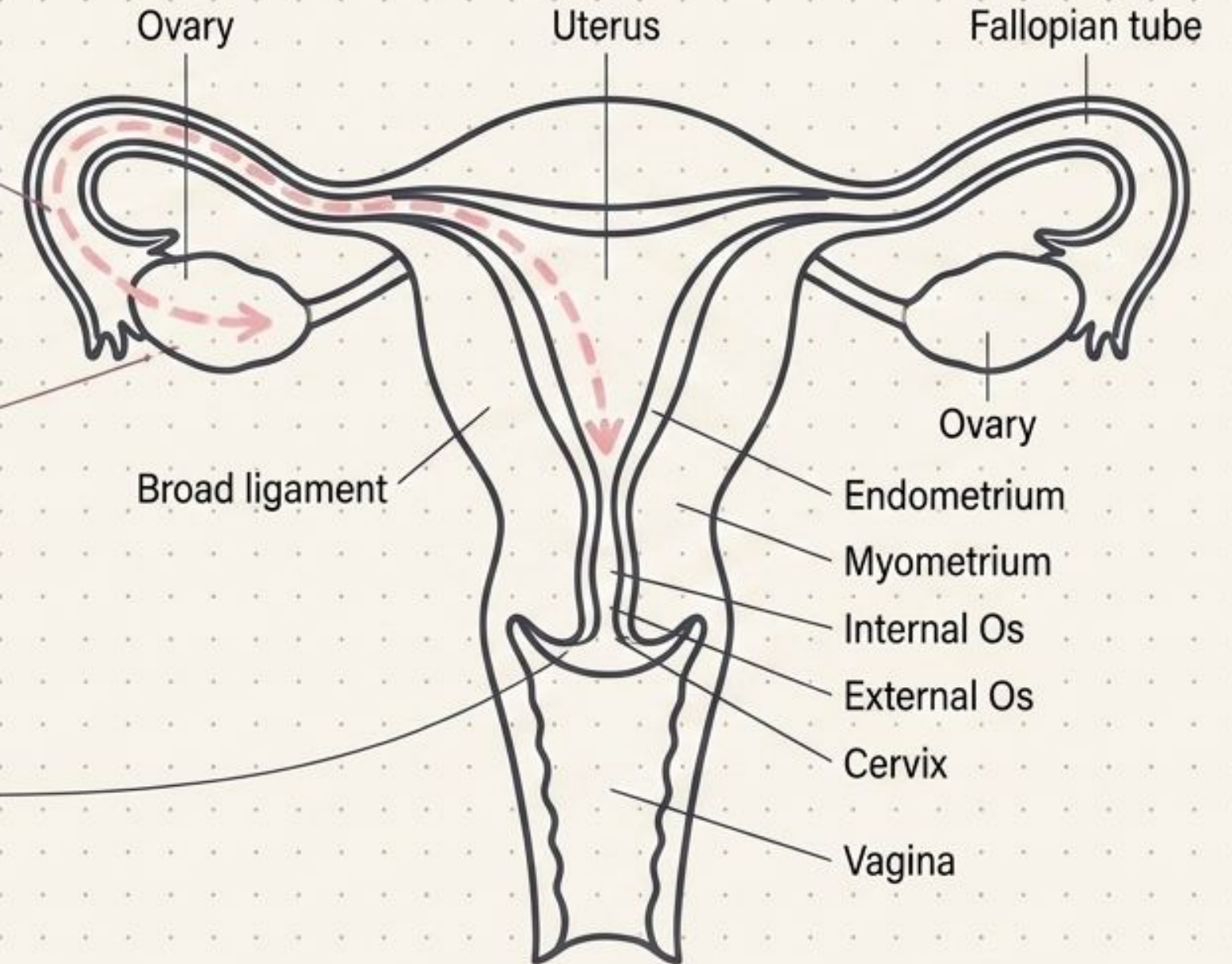
*Fun fact: You have ~1 million immature ova at birth, but only 400-450 mature in a lifetime.*

**Fallopian Tubes:** Transit system carrying egg cells to the uterus (where fertilization actually happens).

**Uterus (Womb):** Incubator organ within the pelvic zone where a fetus is carried.

**Cervix:** Small lower end opening of the uterus.

**Vagina:** Collapsible canal extending backward and upward. Engorges with blood during arousal, triggering mucosal lubricants.



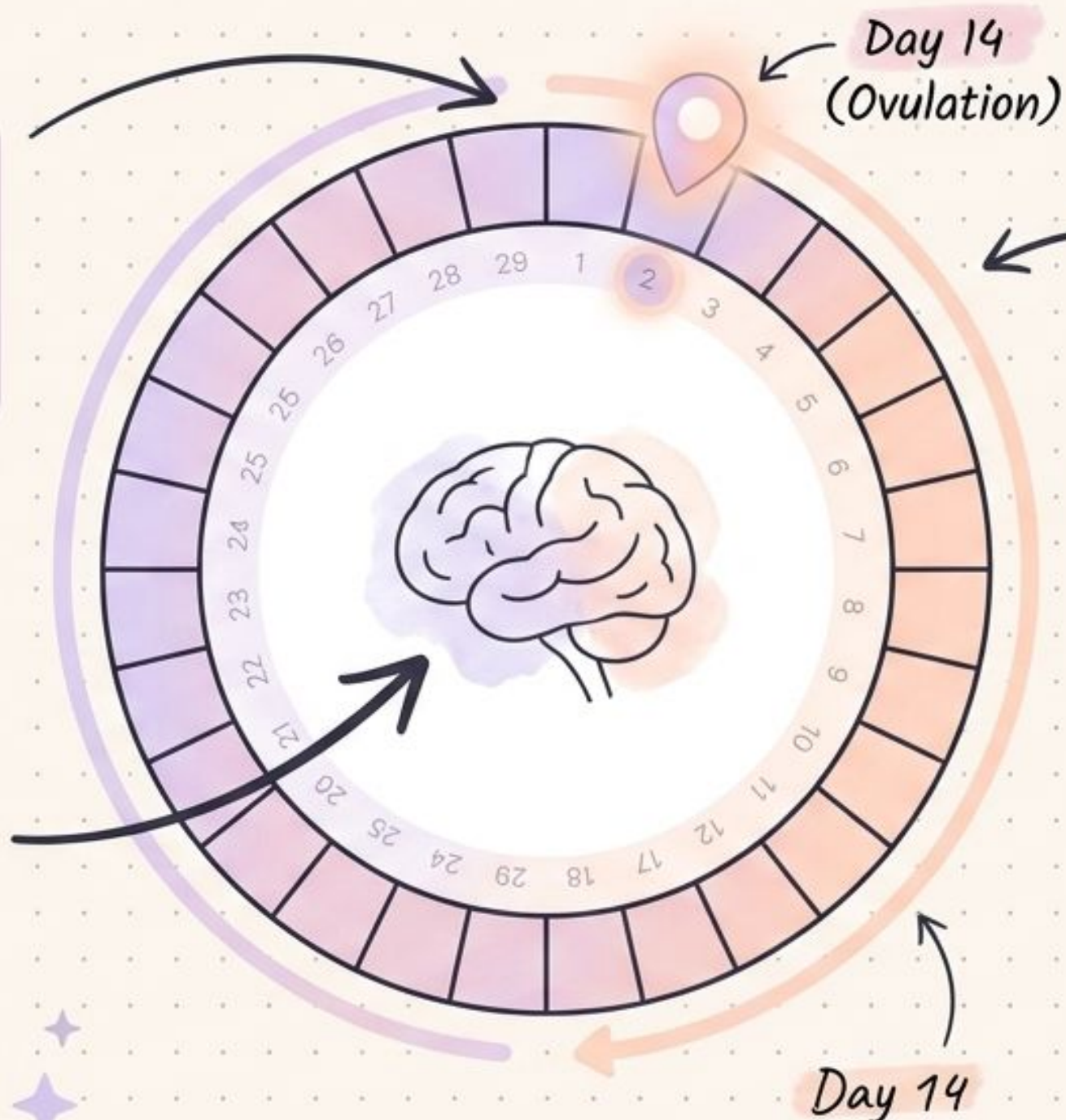
# The Monthly Reset: Menstrual Cycle 101

## Menarche

The very first episode, dropping anywhere between ages 11 to 15.

## The CEO

The overall cycle is governed by the hypothalamus, which acts as the boss monitoring hormone levels in the bloodstream.

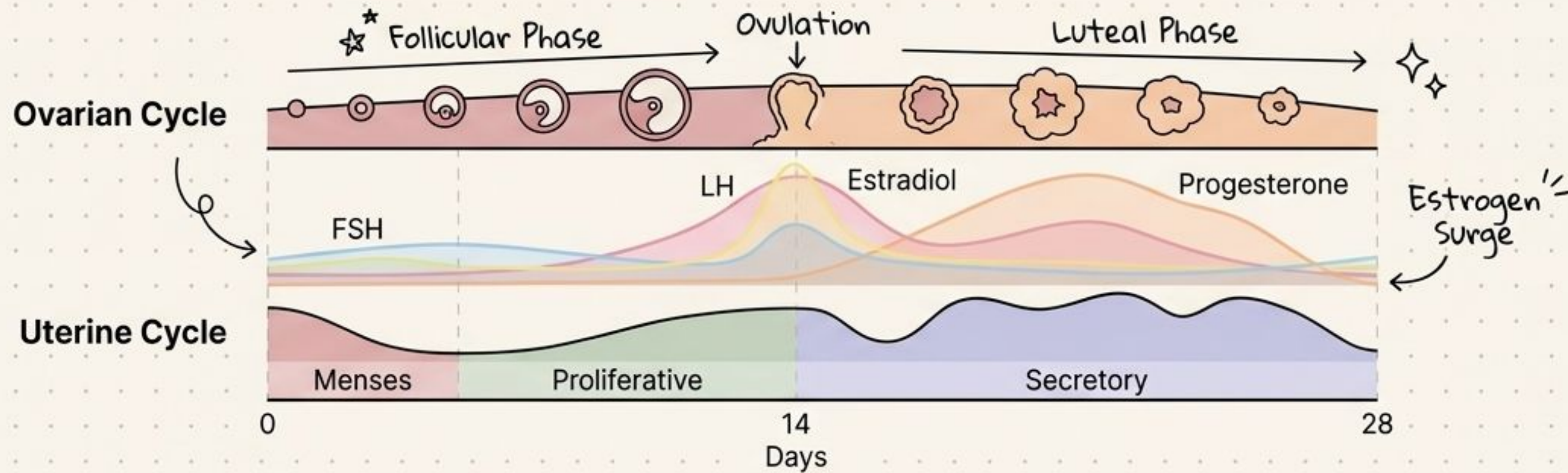


## The Timeline

A standard cycle ranges from 24 to 42 days.

Regardless of length, menstruation begins about 14 days after ovulation (give or take a day or two).

# Hormonal Eras: Tracking the Ovarian and Uterine Cycles



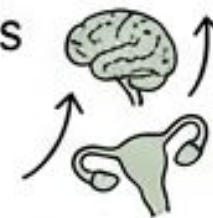
## Menstrual Phase (The Sloughing)

Ovum is unfertilized. Estrogen/progesterone crash to their lowest levels. The uterus throws a fit, endometrium sloughs off, bleeding ensues.



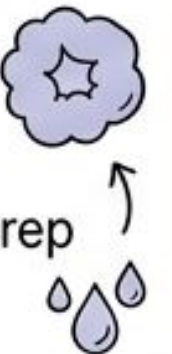
## Proliferative Phase (The Rebuild)

Hypothalamus tells pituitary gland to drop FSH. Estrogen rises, ova matures. Endometrium is repaired and gets highly vascularized.



## Secretory Phase (The Prep)

Pituitary drops LH, triggering ovulation. The leftover follicle becomes the corpus luteum, pumping out progesterone to prep the lining for nutrients.



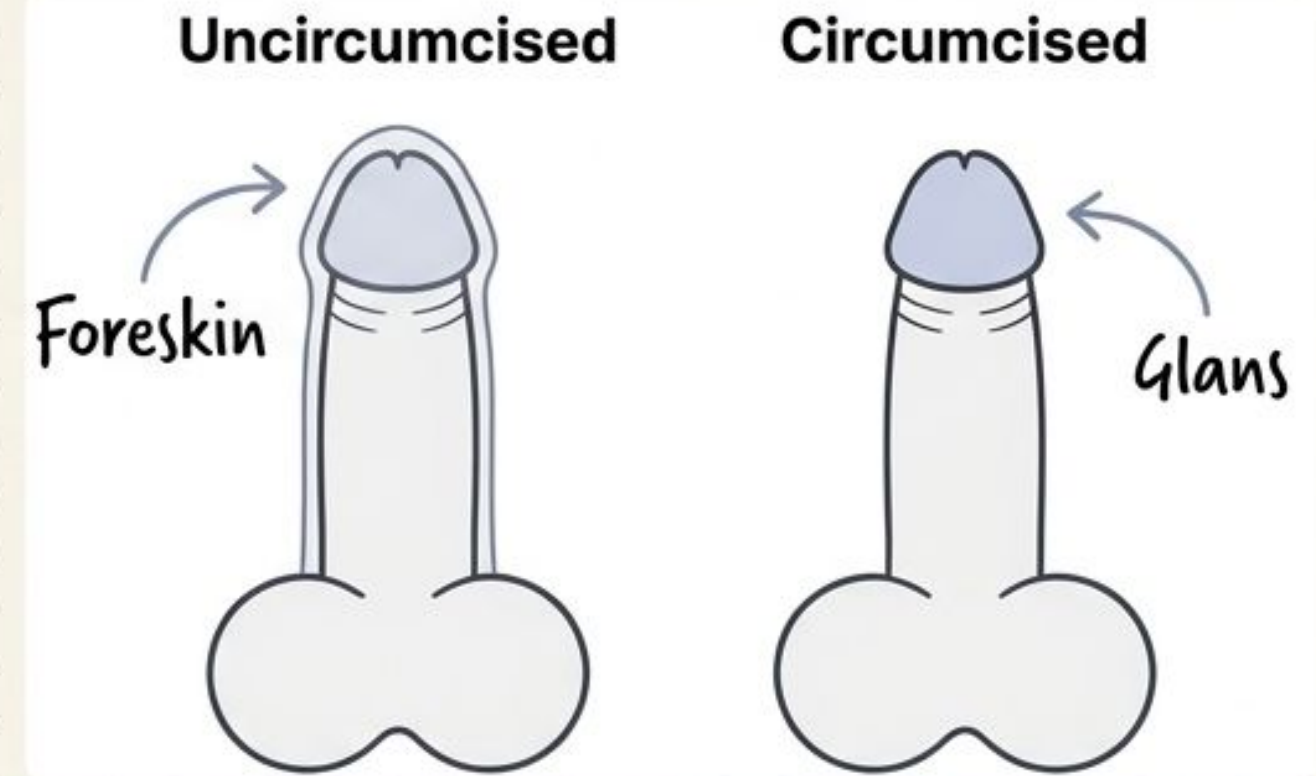
**Glitches in the System:** PMS (hits 80-95% of women), dysmenorrhea (painful cramps from prostaglandins), amenorrhea (disrupted cycle), and menopause (around age 45-50 when ova stop maturing).



# The Outer Hardware: Male External Genitalia

## Inventory Notes

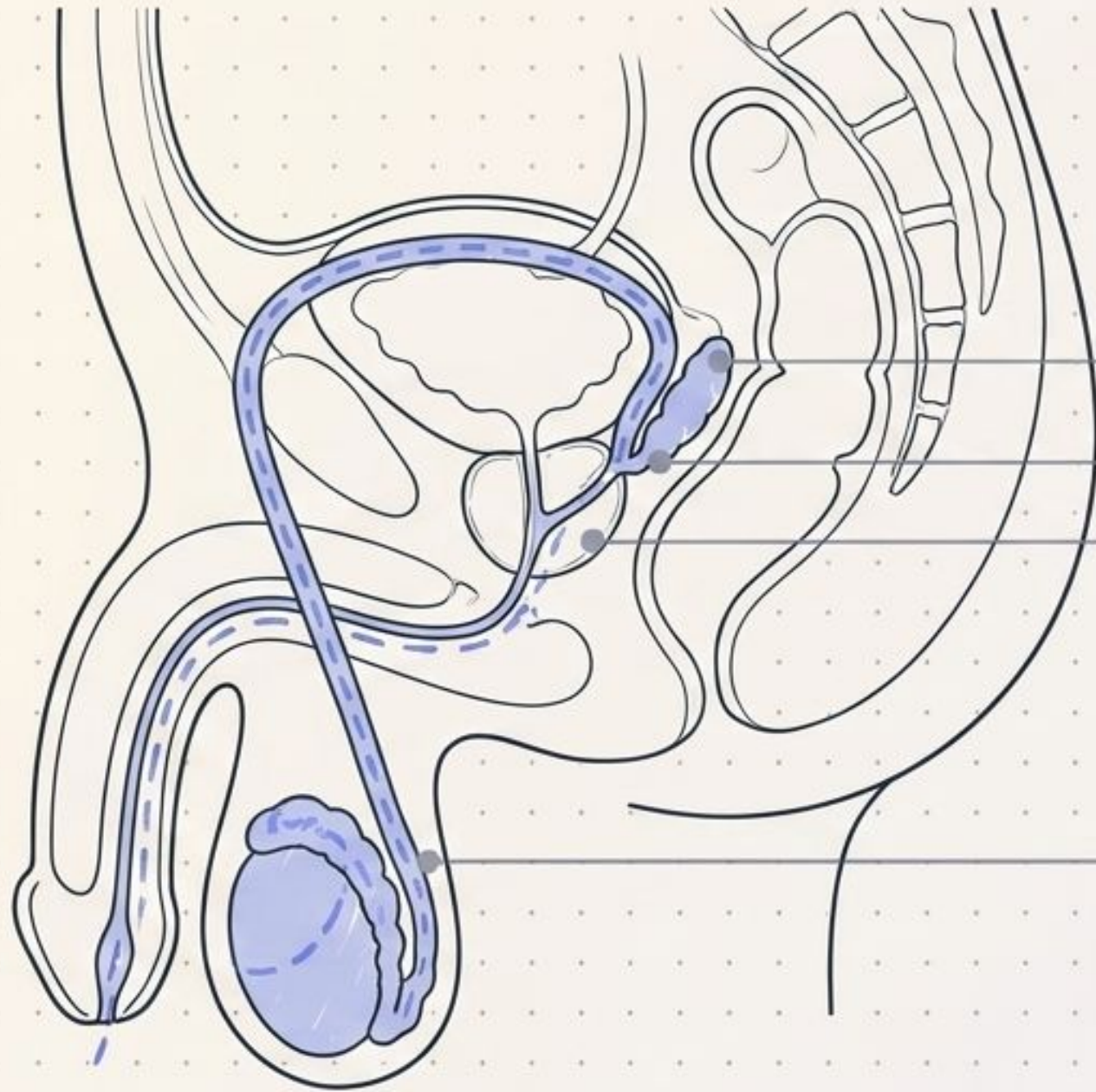
- **Scrotum:** The sac enclosing the two compartments housing the testes.
- **Penis:** Consists of the glans (head), shaft, and root. Surrounded by the spongy body and two cylindrical cavernous bodies (which engorge with blood for erection).
- **Corona & Frenulum:** The rim of the glans, and the thin strip of skin connecting it to the shaft.
- **Urethral Opening:** End of tube connected to bladder for urination and internal structures for semen delivery.
- **Perineum:** Area separating genitalia from the anus (distance is greater in males than females).



### Prepuce (Foreskin)

Covers the head of the penis (removed during circumcision).

# The Assembly Line: VIP Access to the Male Tract



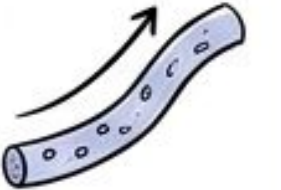
## **Testes:** The production floor

Produces androgen (testosterone) and sperm cells in virtually unlimited quantities over a lifespan.



## **Vas Deferens:** The highway

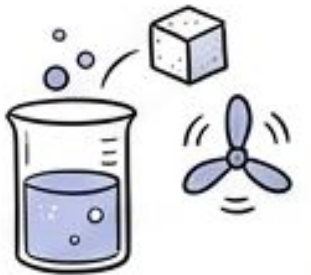
Traveling from the testicle toward the urethra carrying sperm.



## **Seminal Vesicles:** The suppliers

Two glands that produce alkaline fluid rich in fructose sugar (70% of semen volume).

The sugar provides nutrients; the alkaline nature stimulates self-propulsion.

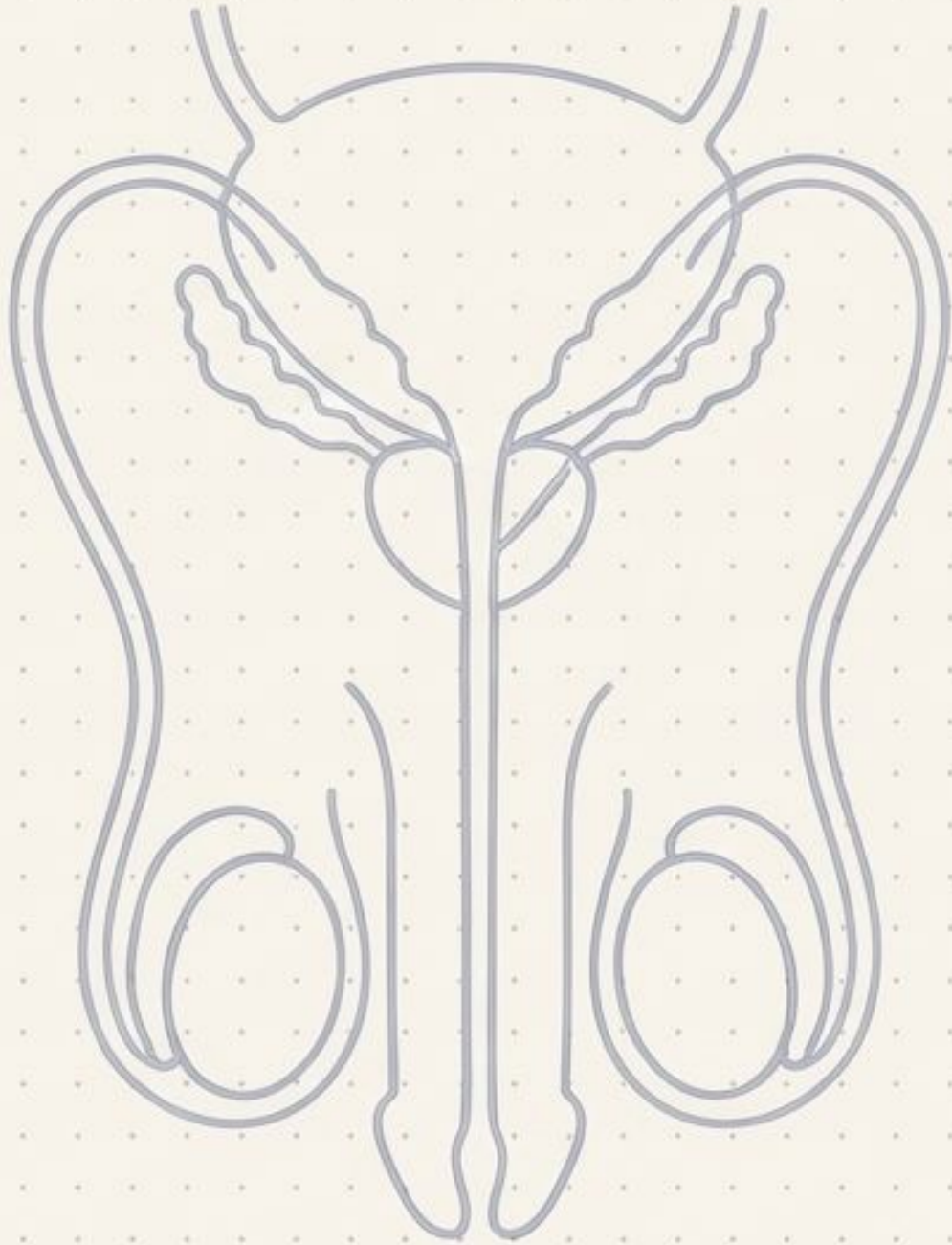


## **Ejaculatory Ducts:** Connect the vas deferens to the urethra.

Connects the vas deferens and seminal vesicle to the urethra.



# The Support System: Glands & Hormones



## Prostate Gland

Produces alkaline secretions (30% of semen volume).

*It understood the assignment: counteracting the acidic environment of the male urethra and female vagina to keep sperm alive.*

## Urethra

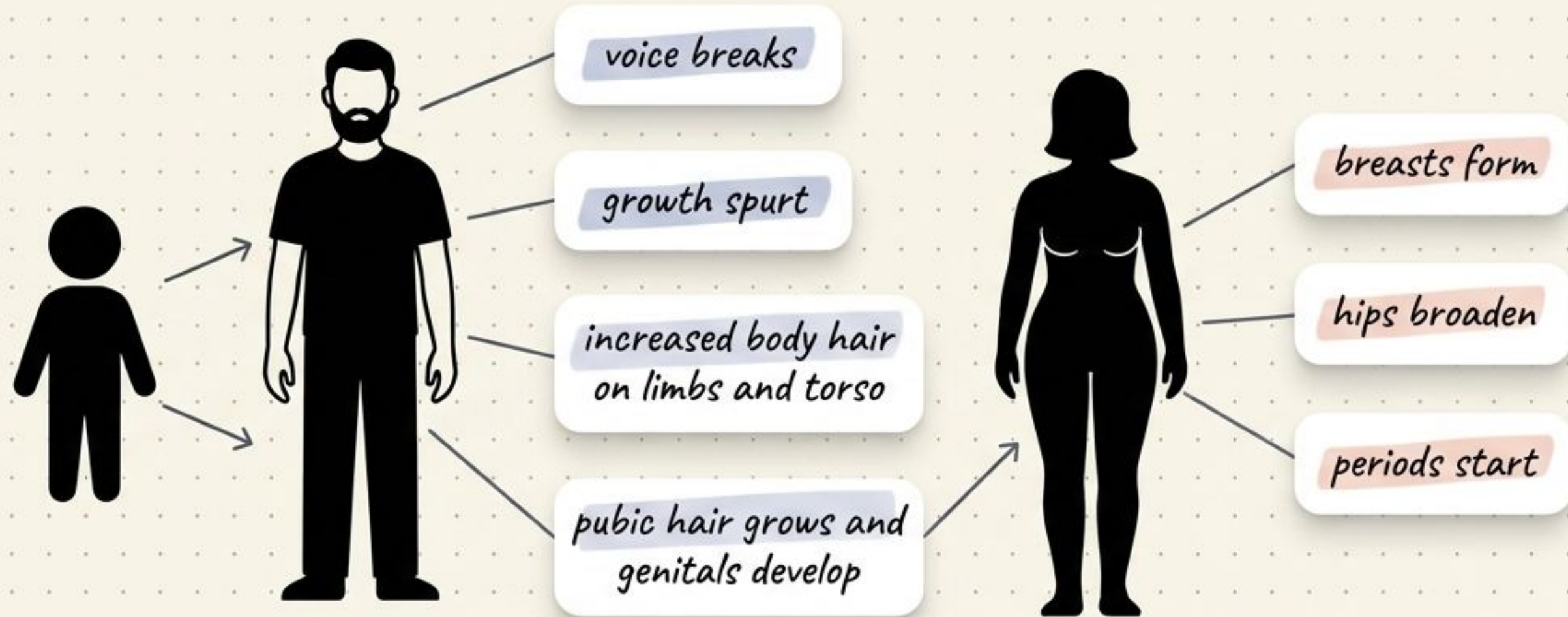
The final tube carrying sperm and semen out of the body.

## Testosterone (The Main Hormone)

Produced mainly by testes (and slightly by adrenal glands). Drives physical appearance, sexual motivation, sperm development, and secondary characteristics during adolescence.



# System Update: Puberty (The Ultimate Glow-Up)



Humans experience dynamic physical and emotional changes from childhood to adulthood.

**The Engine:** The gonads (ovaries in females, testes in males) ramp up the production of germ cells and sex hormones, signaling the final development of primary and accessory organs that support reproduction.

# The Patch Notes: Secondary Sex Characteristics

(Emerging after puberty)

## Female Specs

Widening of hips and pelvis (shifts center of gravity).

Enlargement of breasts (glandular and fatty tissue).

Generally shorter with a greater proportion of body fat.

Two X chromosomes (reduces sex-linked conditions).

Lower mortality rate / longer projected lifespan.

## Male Specs

No monthly cycle.

Broader shoulders and a deeper chest cavity.

Elongation of vocal cords (lower voice).

Taller with a greater proportion of body water.

Proportionately larger heart and lungs.

Single X chromosome (increases colorblindness/hemophilia).

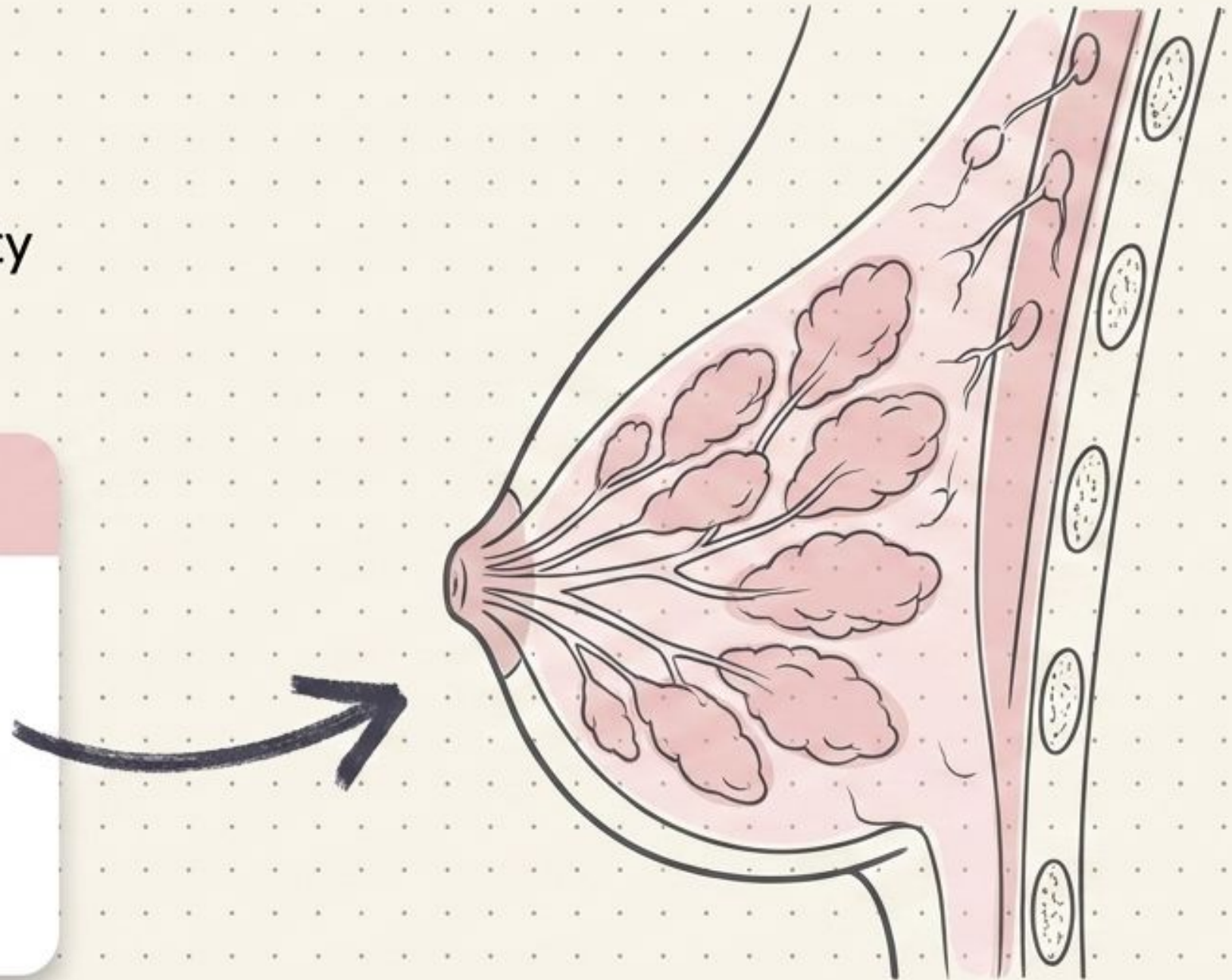
Exposure to higher testosterone = more body/facial hair, but increased baldness.

# Accessory Hardware: Deep Dive on Breasts

At puberty, both glandular and fatty tissues develop considerably.  
(Size differences are primarily due to fatty tissue amounts).

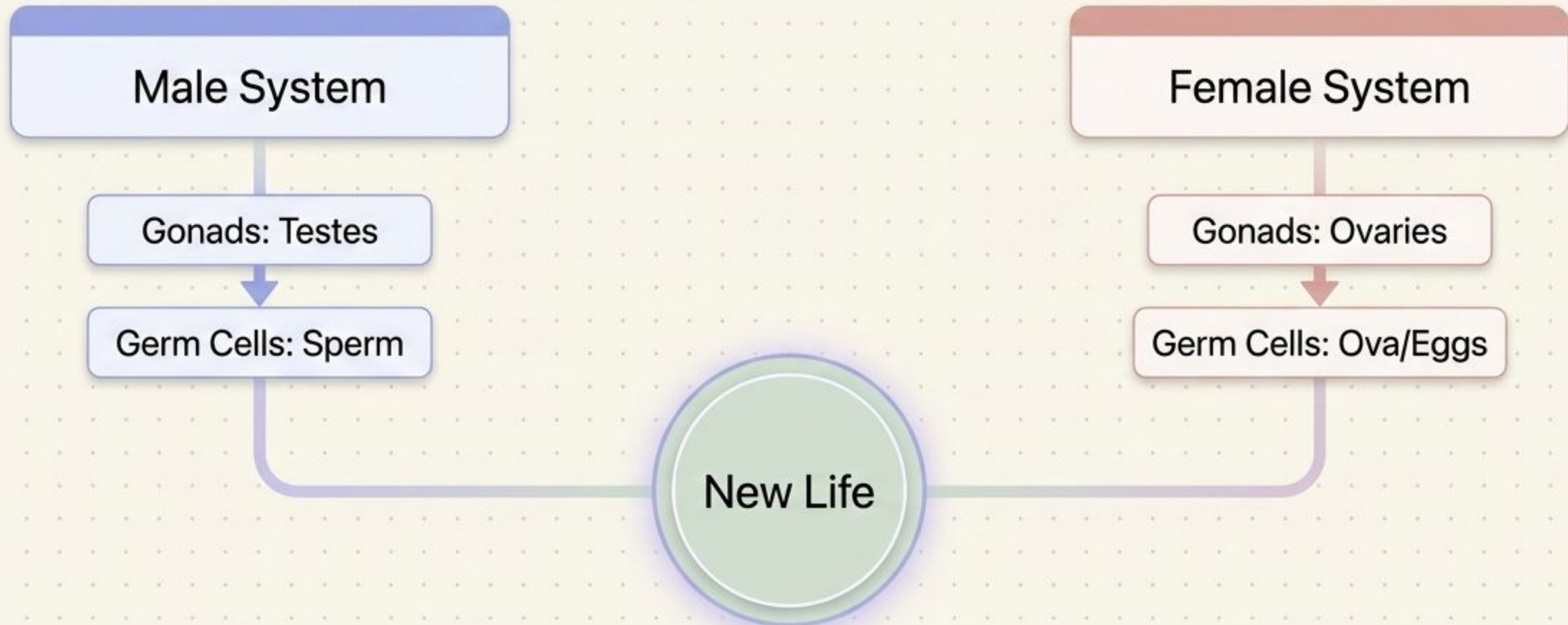
## Glandular Tissue

Highly responsive to sex hormones. It is involved in sexual arousal and produces milk toward the end of pregnancy and after childbirth in response to hormone levels.



# The TL;DR: Two Systems, One Goal

Despite obvious differences in physical appearance, the sexual organs of men and women arise from the same baseline structures and fulfill similar, complementary functions.



*The ultimate objective: living rent-free in the blueprint of human survival.*

# Final Boss Assessment: Check Your Knowledge



## Challenge 1:

Compare and contrast the male and the female genitalia (Hardware check).



## Challenge 2:

Draw a mind map showing the progression of the male and female from birth, through the system update of puberty, and into adulthood.



[PRESS START TO BEGIN]